

James Kirika

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INTRO

Electrical and Electronics Engineering graduate with three-plus years of professional programming experience writing MQL4/5 trading software for MetaQuotes. I'm a fast, self-directed learner — I've picked up C++, C#, Python, and Node.js on the job and through personal projects, with no formal training in any of them beyond documentation and trial and error. Much of my recent work already brushes up against security fundamentals: building encrypted WebSocket (WSS) connections, managing SSL certificates, and implementing token-based authentication for live data bridges between MetaTrader 5 and web servers. My final-year engineering project was an anti-tampering smart meter system built to detect unauthorized access. I'm now looking to turn that instinct into formal cybersecurity skills, bringing the same curiosity and quick-learning approach that's gotten me this far.

WORK EXPERIENCE

Programmer — MetaQuotes

Apr 2023 – Present

Duties & Responsibilities:

At MetaQuotes, I write MQL4/MQL5 programs for trading platforms, including custom indicators, scripts, and Expert Advisors (EAs), and convert trading logic from TradingView's Pine Script into MQL4/5 for clients migrating platforms. I also build C++ and C# DLL bridges that stream live MT5 data to web dashboards and Excel over secure WebSocket (WSS) connections, using SSL certificates and token-based authentication. I collaborate closely with team members to develop, test, and optimize trading algorithms, debug and troubleshoot code, and provide ongoing technical support as client needs evolve.

Key Achievements:

I've developed and deployed custom DLLs, trading indicators, scripts, and Expert Advisors that meaningfully improved trading efficiency and execution speed for clients, including secure real-time bridges authenticated with SSL certificates and tokens. Clients consistently praise the reliability and usability of what I ship, and my ability to deliver it within tight deadlines.

Technician — LED Power Technology

Jan 2022 – Dec 2022

Duties & Responsibilities:

Conduits, electrical installation (wiring and fitting), single-phase and three-phase motor installation and maintenance, installation and maintenance of HVAC systems, CCTV installation and maintenance, and genset servicing.

Key Achievements:

Proved to be an employee who is a quick learner, critical thinker, time-conscious, responsible, and a great team player.

Attachee — KPLC

Jan 2021 – Apr 2021

Duties & Responsibilities:

Low voltage transformer wiring, cable connection on meter bus bars, installation of large power meters and current transformers, power disconnection and reconnection, cable stripping and crimping, meter housing assembly, pole and wall mounting smart meters, and meter reading.

Key Achievements:

Proved to be a disciplined intern who upheld good ethical values both socially and duty-wise. Always punctual and displayed keenness and interest in work.

TECHNICAL PROJECTS

Secure MT5-to-Web Bridge

Designed and built DLLs (C++/C#) that stream live MetaTrader 5 trading data to browser dashboards and Excel in real time — the full pipeline runs from MT5 Expert Advisor, through the DLL, over WebSocket, into a Node.js/FastAPI server, and out to the browser. Implemented WSS with both self-signed and CA-trusted SSL certificates depending on deployment target, added token-based client authentication, and deployed via Cloudflare Tunnel and Render for always-on hosting.

Pine Script to MQL4/5 Conversion

Translate TradingView Pine Script indicators and strategies into MQL4/5 for clients moving to MetaTrader, accounting for differences in execution model, bar timing, and built-in functions between the two platforms.

Notepad Calculator (Python / Kivy)

Built a desktop note-taking app for personal use that combines free-form notes with inline arithmetic — supports variable assignment and reuse within a note, light/dark themes, and undo/redo, designed and coded solo from the UI down to the expression-parsing logic.

Quantitative FX Research

Self-directed research applying Hidden Markov Models to detect volatility regimes across FX pairs, and Kuramoto synchronization theory (borrowed from physics) to study coordinated trader behavior around gold and major USD pairs — including an 8-year walk-forward backtesting and optimization pipeline built in Python.

EDUCATION

Bachelor of Technology in Electrical and Electronics Engineering

Muranga University of Technology • Jan 2017 – Dec 2021

Graduated with second class upper honours, specializing in Power Systems. My final-year project was a smart meter security system: it prevents pre-pay meter bypassing by locking the intake point, granting access only to authorized users, and reporting any tampering attempts to a central database — an early, hands-on introduction to access control and intrusion detection.

KEY SKILLS & COMPETENCIES

Electrical & Technical	Programming & Software
Circuit Design and Implementation	MQL4/MQL5 (Indicators, EAs, Scripts)
Electrical Installation and Assembly	C++ / C# (DLL development)
High Voltage Engineering & Power Line Construction	Python (Kivy, FastAPI, data analysis)
Genset Servicing	Pine Script → MQL4/5 conversion
HVAC Systems Repair	WebSockets, SSL/TLS, secure data pipelines
Three-Phase Networks (Electricity)	Git / GitHub
Computer-Aided Design (Archicad, AutoCAD, Proteus)	Office tools: Word, Excel, PowerPoint

Other skills

Driving license Class B – light vehicles, First Aid and firefighting skills.

LANGUAGES

English — Spoken: Fluent • Written: Fluent

ASSOCIATIONS

Member, MUT Society of Engineering Technology

HOBBIES

Cycling, programming, and travelling.

REFEREES

Please feel free to contact the following in regard to my competence, work ethic, performance, or any other aspect with respect to me:

- Mr. Martin Nderitu — Supervisor, LED Power Technology — Tel: 0781488032
- Prof. Christopher Maina — Dean of School, Engineering, Muranga University of Technology — Tel: 0721791736 — Email: cmmaina@gmail.com